

5.1 Derivation of the Euler–Lagrange Equation

1. Ordinary Calculus: Stationary Means No First-Order Change

Let $g(x)$ be a smooth function. To test whether x is a stationary point, compare $g(x)$ and $g(x + \varepsilon)$, where ε is small.

Taylor expand around x :

$$g(x + \varepsilon) = g(x) + \varepsilon \frac{dg}{dx} + O(\varepsilon^2). \quad (1)$$

Thus,

$$g(x + \varepsilon) - g(x) = \varepsilon \frac{dg}{dx} + O(\varepsilon^2) \quad (2)$$

Therefore,

$$\frac{dg}{dx} = 0 \quad (3)$$

means the change has **no first-order term**. This is the precise meaning of being **locally flat**.

2. Functionals: The Same Idea in Function Space

Now consider the functional

$$F[y] = \int_{x_1}^{x_2} f(x, y, y') dx, \quad (4)$$

with fixed boundary values $y(x_1)$ and $y(x_2)$.

To test whether $y(x)$ is stationary, perturb it:

$$y_\varepsilon(x) = y(x) + \varepsilon \eta(x),$$

where

$$\eta(x_1) = \eta(x_2) = 0.$$

The derivative also changes:

$$y'_\varepsilon(x) = y'(x) + \varepsilon \eta'(x).$$

So

$$F[y + \varepsilon \eta] = \int_{x_1}^{x_2} f(x, y + \varepsilon \eta, y' + \varepsilon \eta') dx. \quad (5)$$

3. Taylor Expand Around y

At each x , expand the integrand around the original values $y(x)$ and $y'(x)$:

$$f(x, y + \varepsilon\eta, y' + \varepsilon\eta') = f(x, y, y') + \varepsilon \left(\frac{\partial f}{\partial y} \eta + \frac{\partial f}{\partial y'} \eta' \right) + O(\varepsilon^2). \quad (6)$$

Therefore,

$$F[y + \varepsilon\eta] = F[y] + \varepsilon \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y} \eta + \frac{\partial f}{\partial y'} \eta' \right) dx + O(\varepsilon^2). \quad (7)$$

The coefficient of ε is the first variation:

$$\delta F[y; \eta] = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y} \eta + \frac{\partial f}{\partial y'} \eta' \right) dx. \quad (8)$$

A stationary function satisfies $\delta F[y; \eta] = 0$ for every admissible η .

4. Integration by Parts

The term containing η' is integrated by parts:

$$\int_{x_1}^{x_2} \frac{\partial f}{\partial y'} \eta' dx = \left. \frac{\partial f}{\partial y'} \eta \right|_{x_1}^{x_2} - \int_{x_1}^{x_2} \eta \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) dx. \quad (9)$$

Since $\eta(x_1) = \eta(x_2) = 0$, the boundary term vanishes. Hence

$$\delta F[y; \eta] = \int_{x_1}^{x_2} \eta \left[\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \right] dx. \quad (10)$$

5. Euler-Lagrange Equation

For stationarity, this must vanish for every η :

$$\int_{x_1}^{x_2} \eta \left[\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \right] dx = 0. \quad (11)$$

By the fundamental lemma of the calculus of variations,

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0. \quad (12)$$

Therefore,

$$\boxed{\frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) - \frac{\partial f}{\partial y} = 0} \quad (13)$$

is the Euler–Lagrange equation.

Key Point

The derivation is just Taylor expansion in function space:

$$F[y + \varepsilon\eta] = F[y] + \varepsilon\delta F[y; \eta] + O(\varepsilon^2).$$

Stationary means the first-order term vanishes:

$$\delta F[y; \eta] = 0.$$